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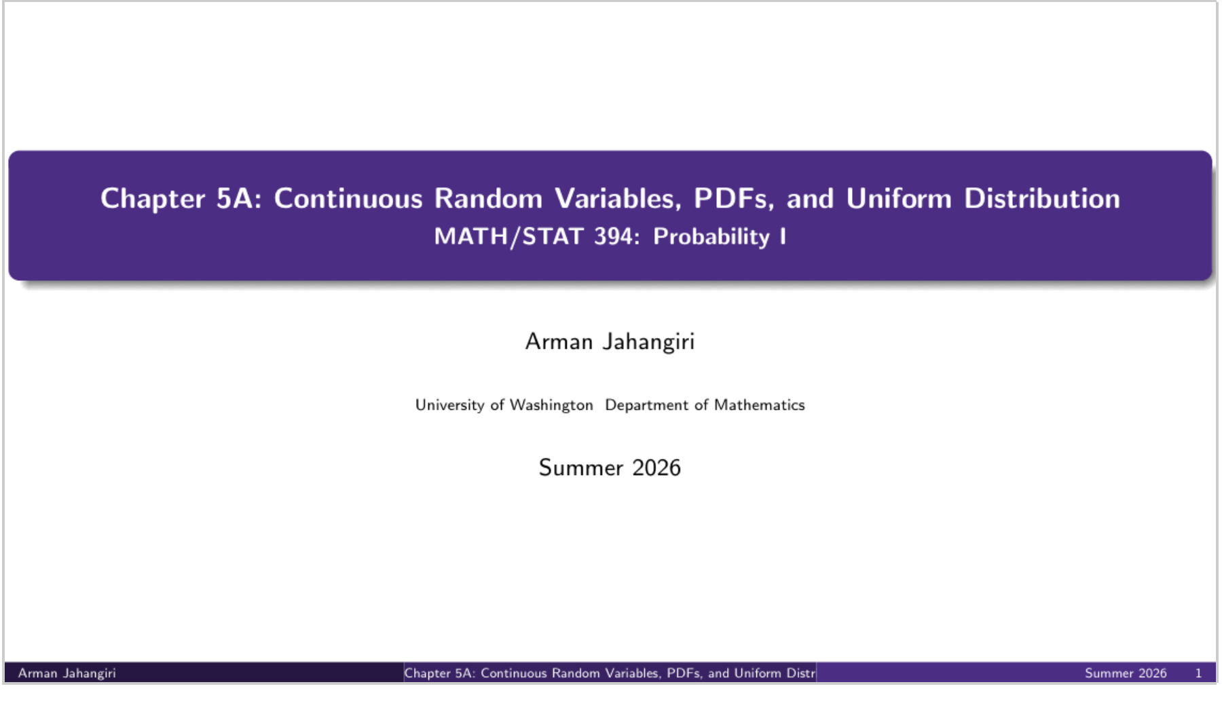

Chapter 5A: Continuous Random Variables, PDFs, and Uniform Distribution

MATH/STAT 394: Probability I

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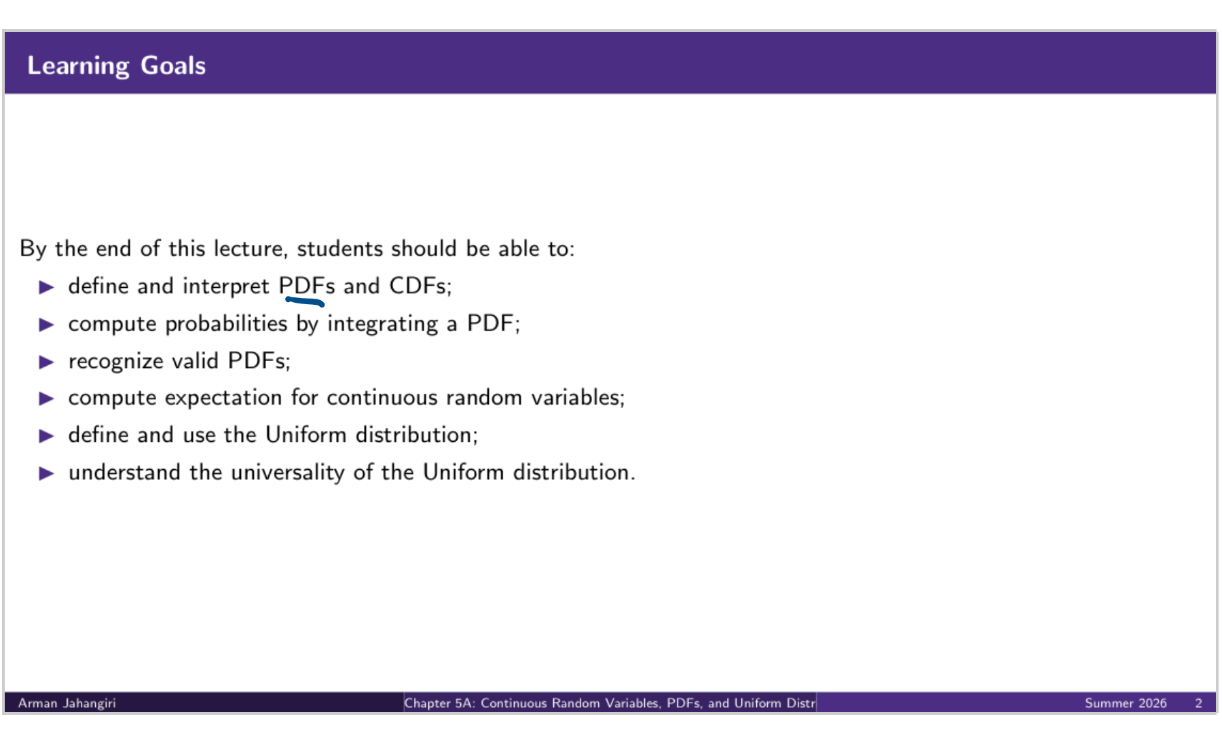
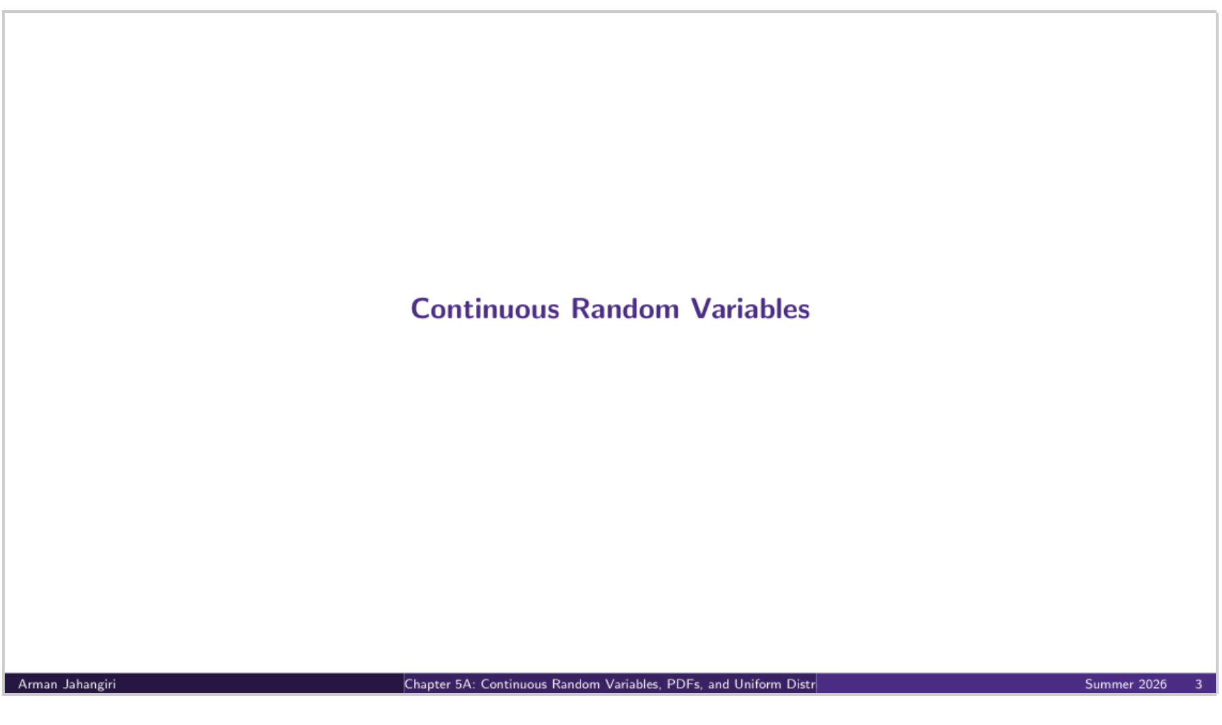


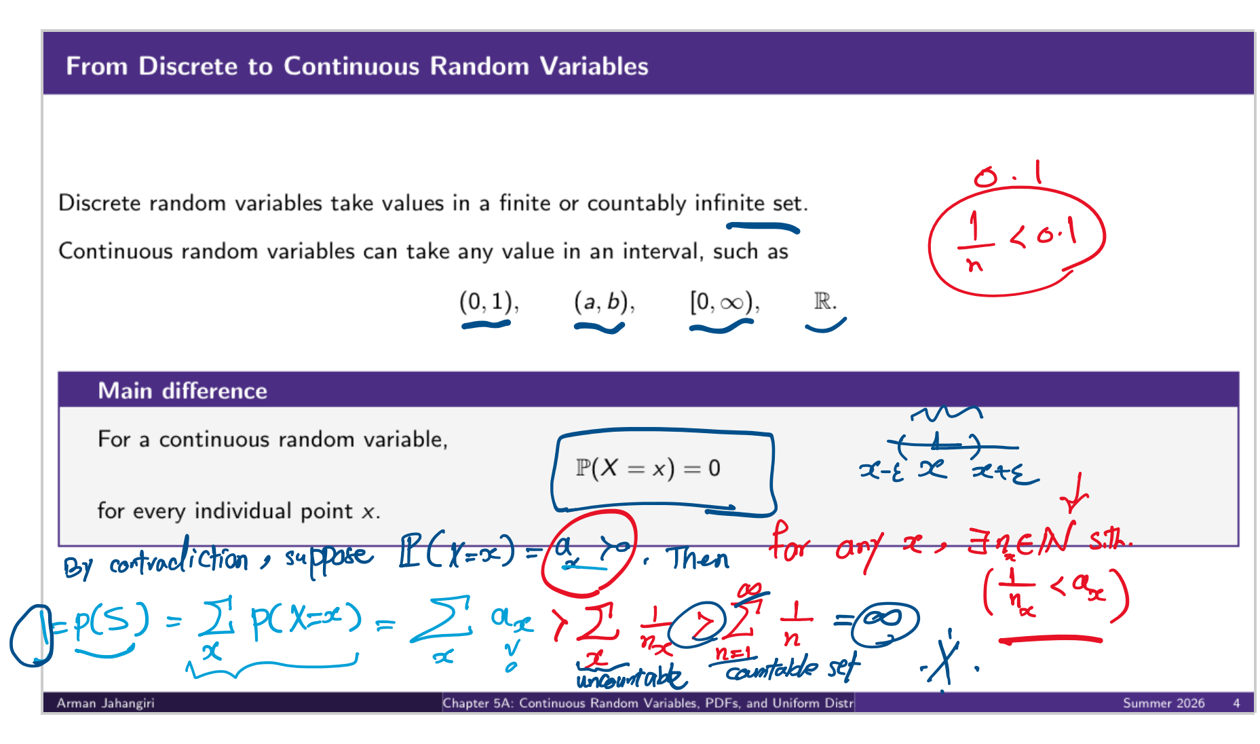
Learning Goals

By the end of this lecture, students should be able to:

- define and interpret PDFs and CDFs;
- compute probabilities by integrating a PDF;
- recognize valid PDFs;
- compute expectation for continuous random variables;
- define and use the Uniform distribution;
- understand the universality of the Uniform distribution.



Continuous Random Variables



From Discrete to Continuous Random Variables

Discrete random variables take values in a finite or countably infinite set.

Continuous random variables can take any value in an interval, such as

$$(0,1), \quad (a,b), \quad [0,\infty), \quad \mathbb{R}.$$

0.1

$\frac{1}{n} < 0.1$

Main difference

For a continuous random variable,

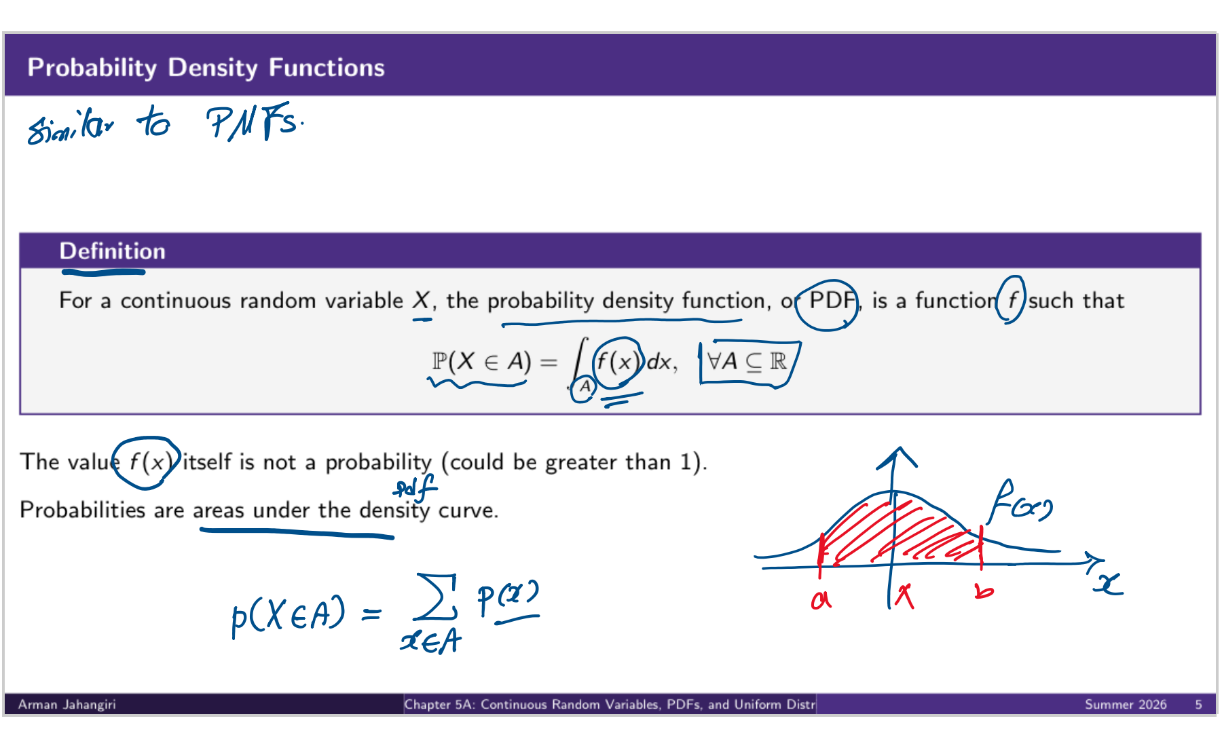
$$\boxed{\mathbb{P}(X=x)=0}$$

for every individual point x.

$x-\varepsilon \quad x \quad x+\varepsilon$

By contradiction, suppose $\mathbb{P}(X=x) = a_x > 0$. Then for any x, $\exists\, n_x \in \mathbb{N}$ s.th. $\left(\frac{1}{n_x} < a_x\right)$

$$1 = P(S) = \sum_x P(X=x) = \sum_x \underset{\underset{0}{\vee}}{a_x} > \sum_{\underset{\text{uncountable}}{x}} \frac{1}{n_x} \geq \sum_{\underset{\text{countable set}}{n=1}}^{\infty} \frac{1}{n} = \infty \quad X$$



Probability Density Functions

Similar to PMFs.

Definition

For a continuous random variable X, the probability density function, or PDF, is a function f such that

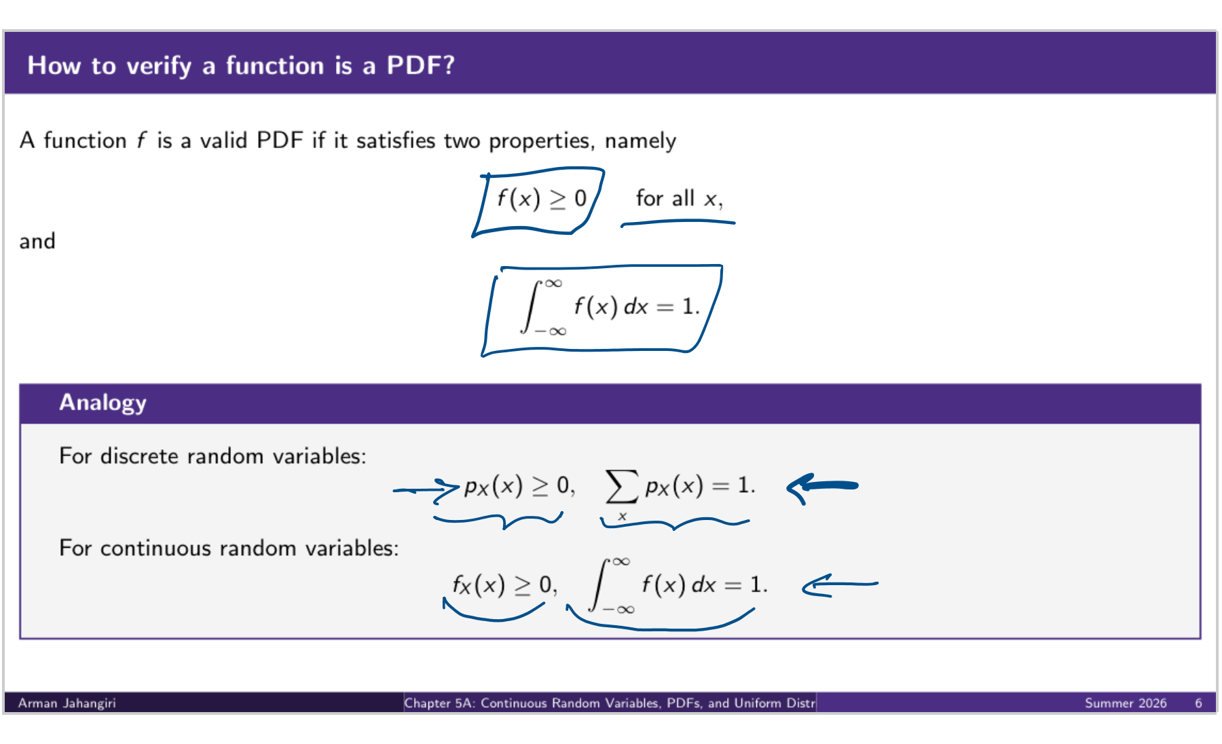
$$\mathbb{P}(X \in A) = \int_A f(x)\,dx, \quad \forall A \subseteq \mathbb{R}$$

The value $f(x)$ itself is not a probability (could be greater than 1).

Probabilities are areas under the density curve.

pdf

$$p(X \in A) = \sum_{x \in A} p(x)$$



How to verify a function is a PDF?

A function f is a valid PDF if it satisfies two properties, namely

$$\boxed{f(x) \geq 0} \quad \text{for all } x,$$

and

$$\boxed{\int_{-\infty}^{\infty} f(x)\,dx = 1.}$$

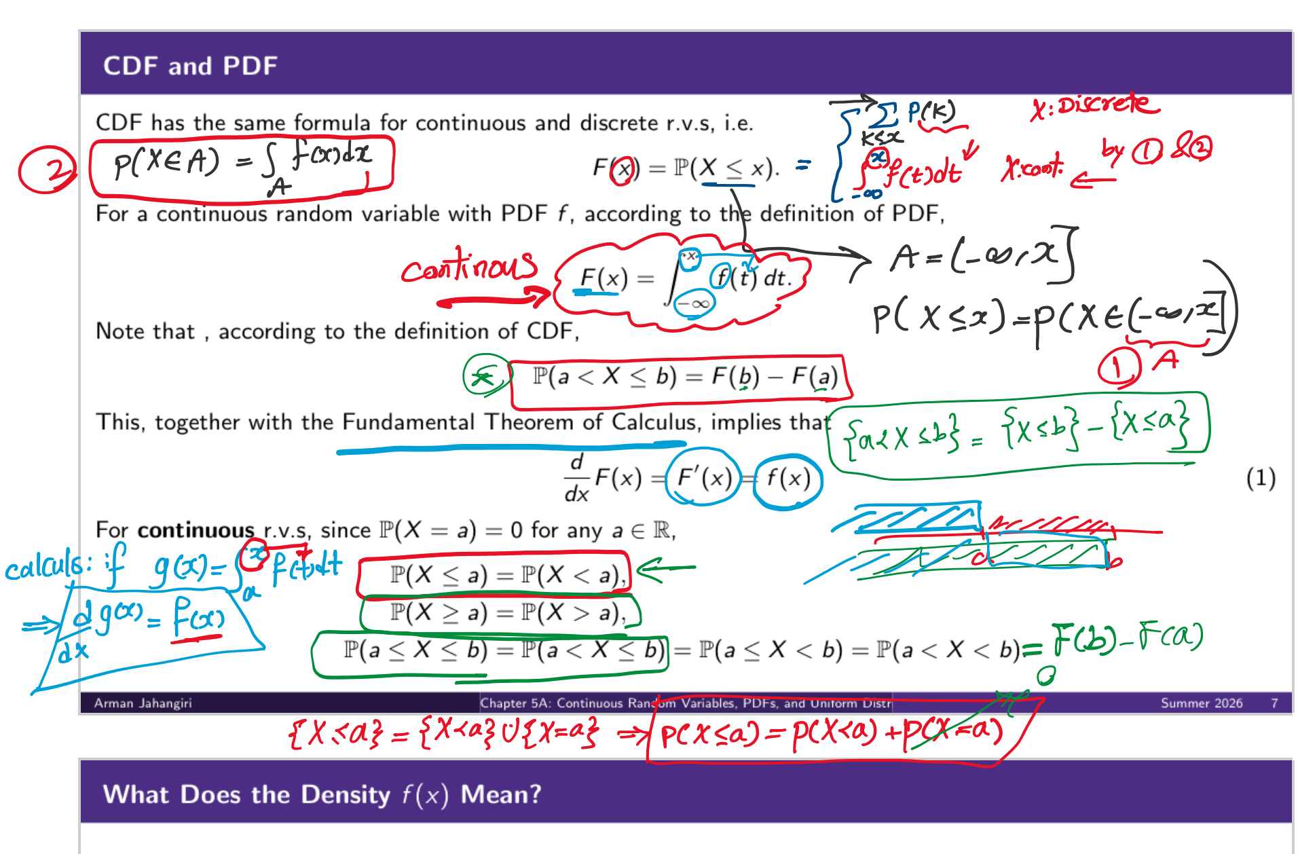
Analogy

For discrete random variables:

$$\to p_X(x) \geq 0, \quad \sum_x p_X(x) = 1. \leftarrow$$

For continuous random variables:

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x)\,dx = 1. \leftarrow$$



CDF and PDF

CDF has the same formula for continuous and discrete r.v.s, i.e.

$$F(x) = \mathbb{P}(X \leq x). = \begin{cases} \sum_{k \leq x} P(k) & X\text{: Discrete} \\ \int_{-\infty}^{x} f(t)\,dt & X\text{: cont.} \end{cases} \quad \text{by ① \& ②}$$

② $\boxed{P(X \in A) = \int_A f(x)\,dx}$

For a continuous random variable with PDF f, according to the definition of PDF,

continous $\to$

$$F(x) = \int_{-\infty}^{x} f(t)\,dt. \to A = (-\infty, x]$$

$$P(X \leq x) = P(X \in (-\infty, x]) \quad ① A$$

Note that , according to the definition of CDF,

$$✱ \; \boxed{\mathbb{P}(a < X \leq b) = F(b) - F(a)}$$

$$\{a < X \leq b\} = \{X \leq b\} - \{X \leq a\}$$

This, together with the Fundamental Theorem of Calculus, implies that

$$\frac{d}{dx}F(x) = F'(x) = f(x) \tag{1}$$

For **continuous** r.v.s, since $\mathbb{P}(X=a)=0$ for any $a \in \mathbb{R}$,

calculs: if $g(x) = \int_a^x f(t)\,dt$ ③ $\Rightarrow \frac{d}{dx} g(x) = f(x)$

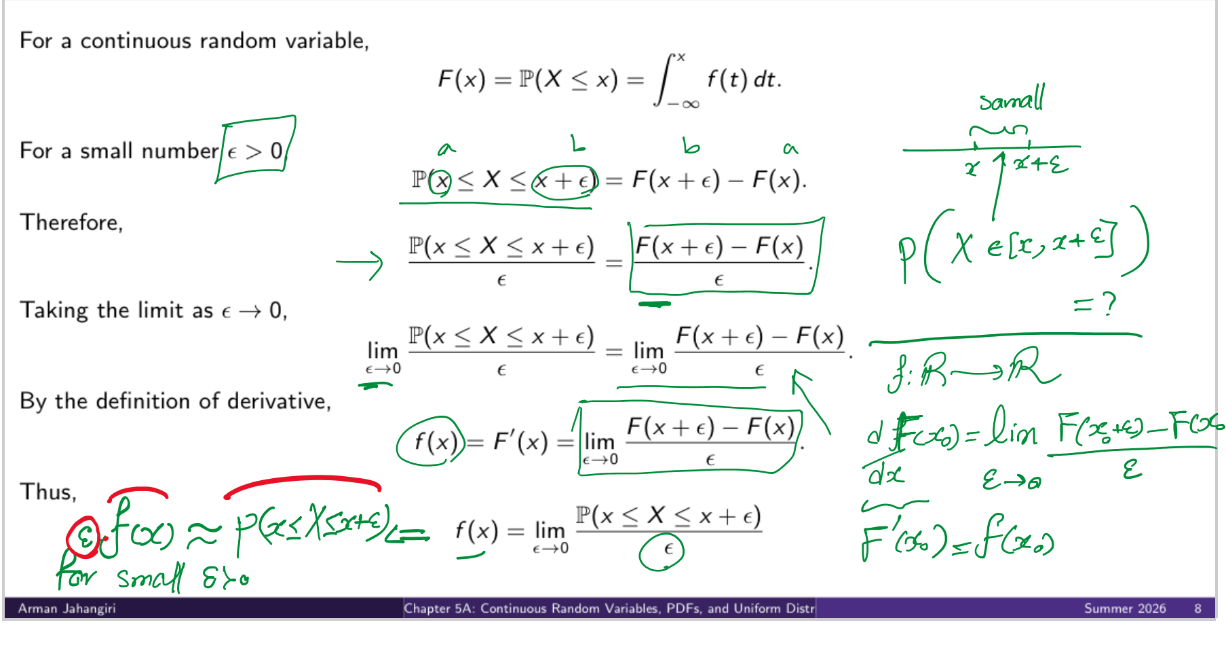
$$\boxed{\mathbb{P}(X \leq a) = \mathbb{P}(X < a),} \leftarrow$$

$$\mathbb{P}(X \geq a) = \mathbb{P}(X > a),$$

$$\mathbb{P}(a \leq X \leq b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a < X < b) = F(b) - F(a)$$

$$\{X \leq a\} = \{X < a\} \cup \{X = a\} \Rightarrow \boxed{P(X \leq a) = P(X < a) + \cancel{P(X=a)}}$$

0



What Does the Density $f(x)$ Mean?

For a continuous random variable,

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f(t)\,dt.$$

For a small number $\boxed{\epsilon > 0}$

$$\mathbb{P}(\overset{a}{x} \leq X \leq \overset{b}{x+\epsilon}) = F(\overset{b}{x+\epsilon}) - F(\overset{a}{x}).$$

Therefore,

$$\to \frac{\mathbb{P}(x \leq X \leq x+\epsilon)}{\epsilon} = \boxed{\frac{F(x+\epsilon) - F(x)}{\epsilon}}.$$

Taking the limit as $\epsilon \to 0$,

$$\lim_{\epsilon \to 0} \frac{\mathbb{P}(x \leq X \leq x+\epsilon)}{\epsilon} = \lim_{\epsilon \to 0} \frac{F(x+\epsilon) - F(x)}{\epsilon}.$$

By the definition of derivative,

$$f(x) = F'(x) = \boxed{\lim_{\epsilon \to 0} \frac{F(x+\epsilon) - F(x)}{\epsilon}}.$$

Thus,

$$\varepsilon \cdot f(x) \approx P(x \leq X \leq x+\varepsilon) \Leftarrow f(x) = \lim_{\epsilon \to 0} \frac{\mathbb{P}(x \leq X \leq x+\epsilon)}{\epsilon}$$

for small $\varepsilon > 0$

small

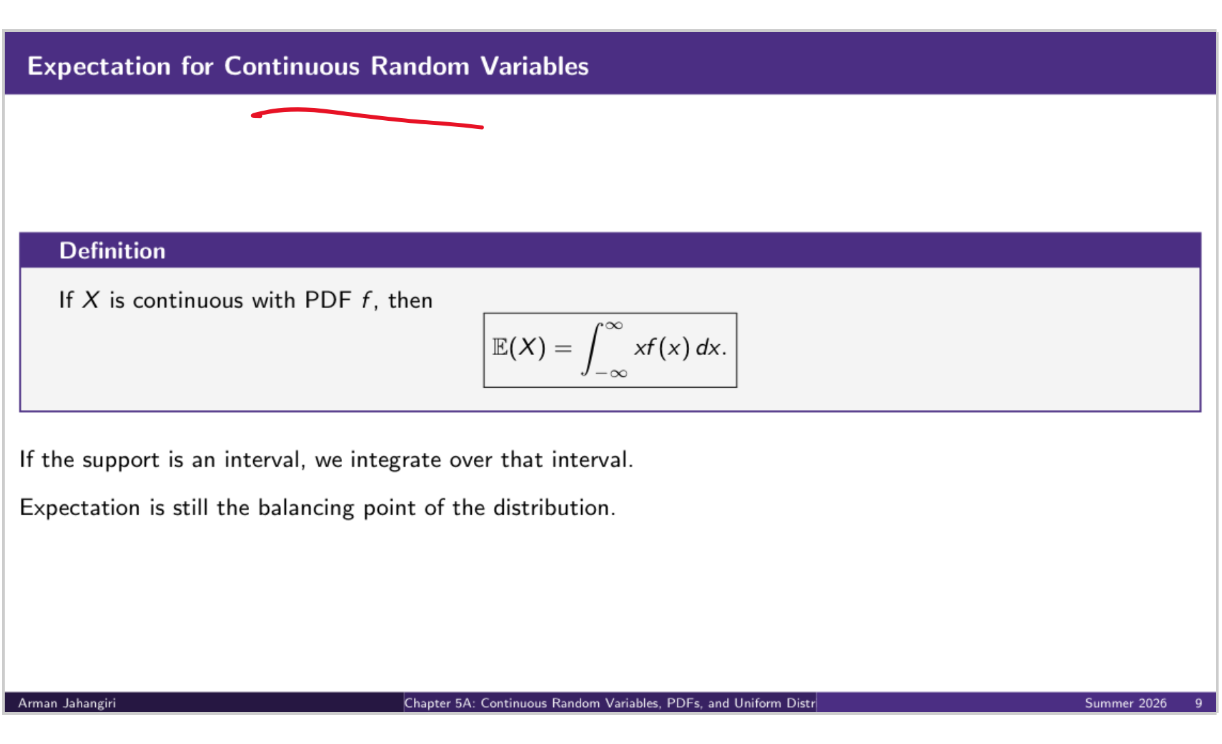
$x \quad x+\varepsilon$

$$P(X \in [x, x+\varepsilon]) = ?$$

$$f: \mathbb{R} \to \mathbb{R}$$

$$\frac{d F(x_0)}{dx} = \lim_{\varepsilon \to 0} \frac{F(x_0+\varepsilon) - F(x_0)}{\varepsilon}$$

$$F'(x_0) = f(x_0)$$



Expectation for Continuous Random Variables

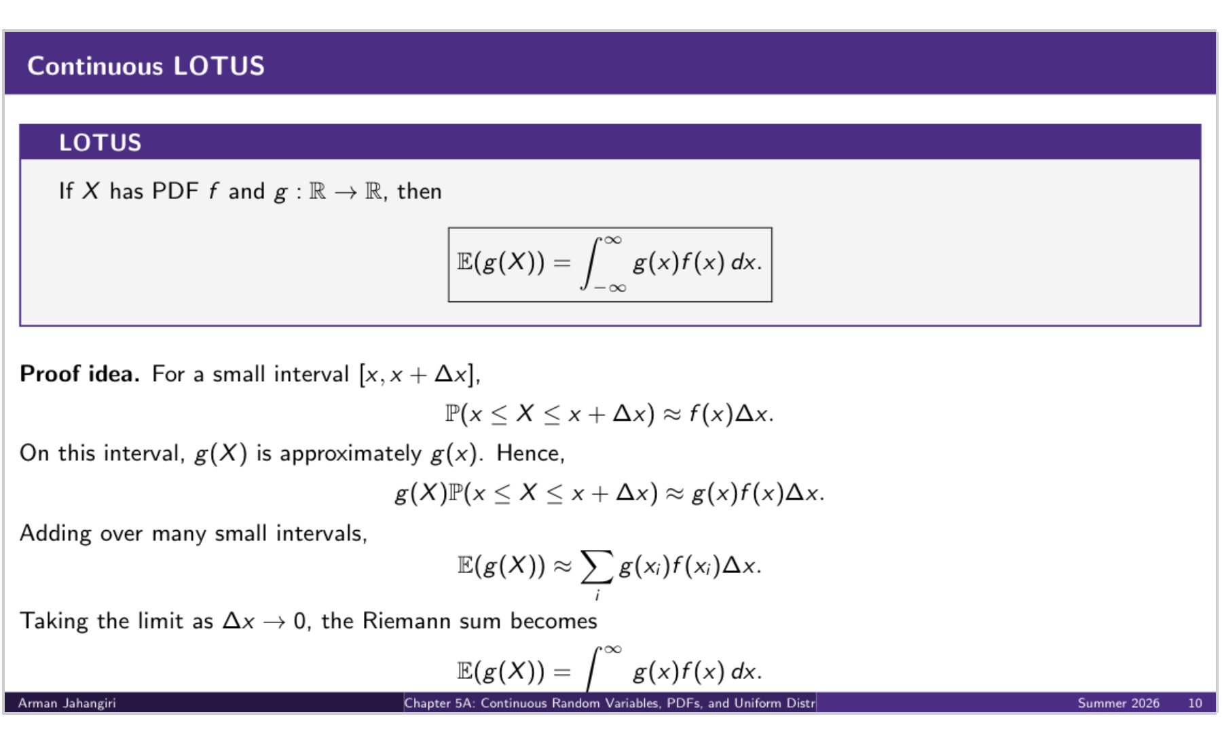
Definition

If X is continuous with PDF f, then

$$\boxed{\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x)\,dx.}$$

If the support is an interval, we integrate over that interval.

Expectation is still the balancing point of the distribution.



Continuous LOTUS

LOTUS

If X has PDF f and $g : \mathbb{R} \to \mathbb{R}$, then

$$\boxed{\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f(x)\,dx.}$$

Proof idea. For a small interval $[x, x+\Delta x]$,

$$\mathbb{P}(x \leq X \leq x+\Delta x) \approx f(x)\Delta x.$$

On this interval, $g(X)$ is approximately $g(x)$. Hence,

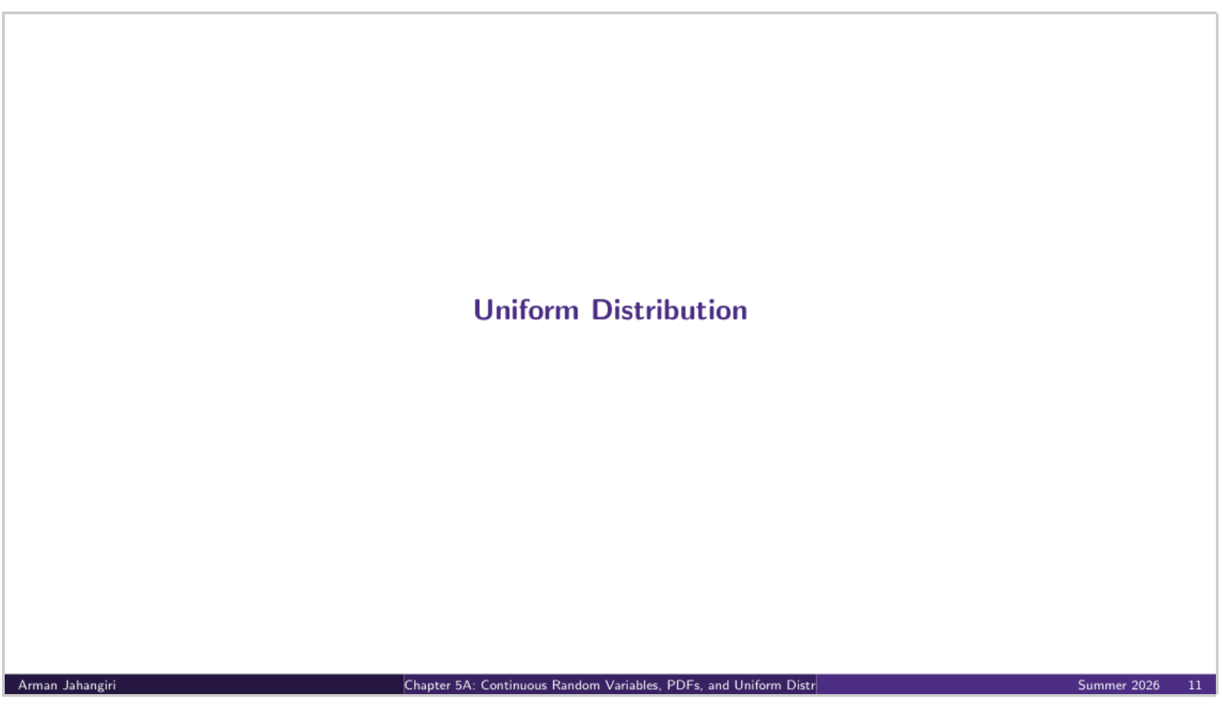
$$g(X)\mathbb{P}(x \leq X \leq x+\Delta x) \approx g(x) f(x)\Delta x.$$

Adding over many small intervals,

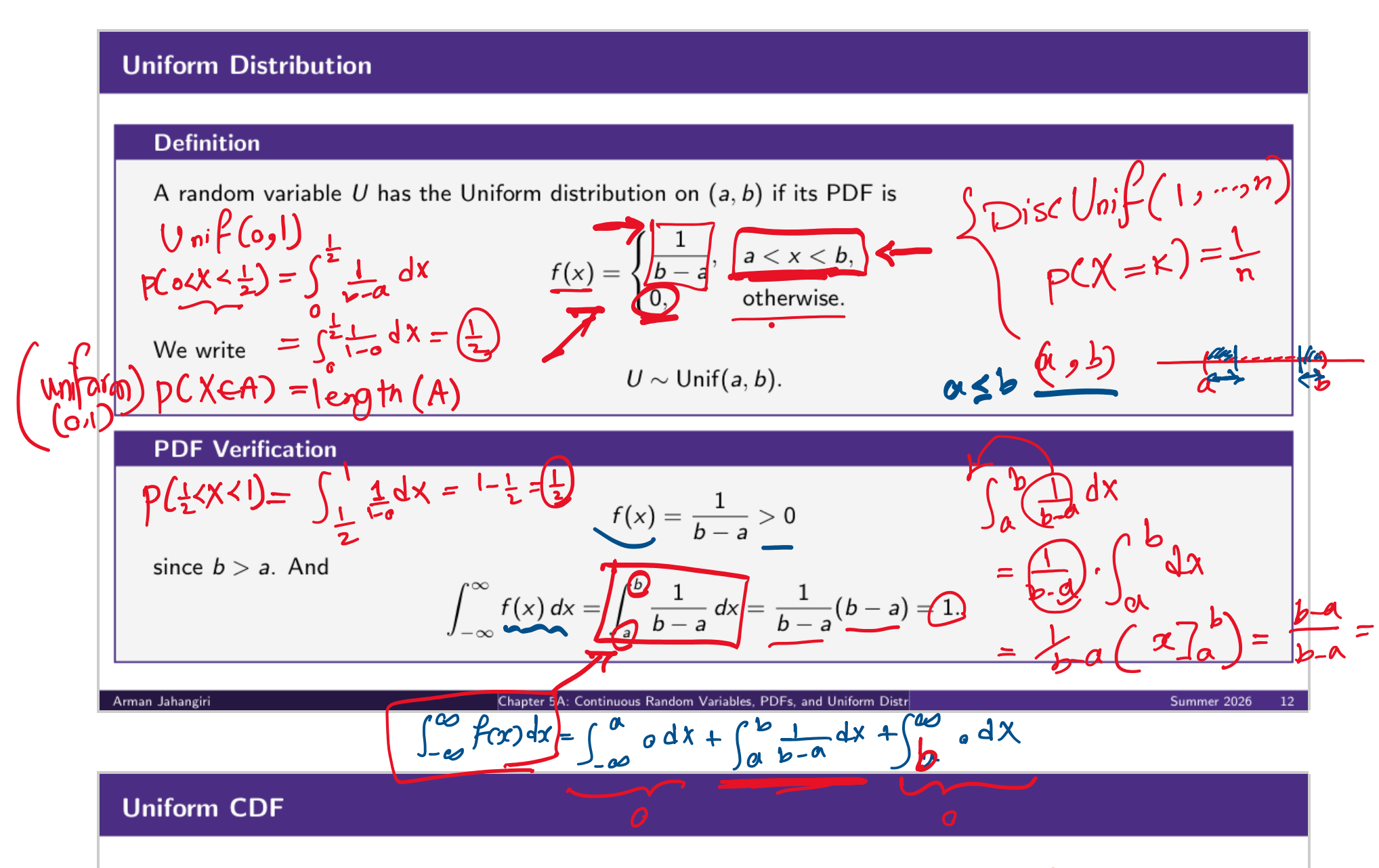
$$\mathbb{E}(g(X)) \approx \sum_i g(x_i) f(x_i)\Delta x.$$

Taking the limit as $\Delta x \to 0$, the Riemann sum becomes

$$\mathbb{E}(g(X)) = \int^{\infty} g(x) f(x)\,dx.$$



Uniform Distribution



Uniform Distribution

Definition

A random variable U has the Uniform distribution on (a,b) if its PDF is

$$f(x) = \begin{cases} \frac{1}{b-a}, & \boxed{a < x < b,} \leftarrow \\ 0, & \text{otherwise.} \end{cases}$$

Unif(0,1)

$$P(0 < X < \tfrac{1}{2}) = \int_0^{\frac{1}{2}} \frac{1}{b-a}\,dx = \int_0^{\frac{1}{2}} \frac{1}{1-0}\,dx = \frac{1}{2}$$

We write

$$U \sim \text{Unif}(a,b).$$

(uniform (0,1)) $p(X \in A) = \text{length}(A)$

$$\left\{ \text{Disc Unif}(1, \dots, n) \quad p(X=x) = \frac{1}{n} \right.$$

$a \leq b \quad (a, b)$

PDF Verification

$$P(\tfrac{1}{2} < X < 1) = \int_{\frac{1}{2}}^{1} \frac{1}{1-0}\,dx = 1 - \frac{1}{2} = \frac{1}{2}$$

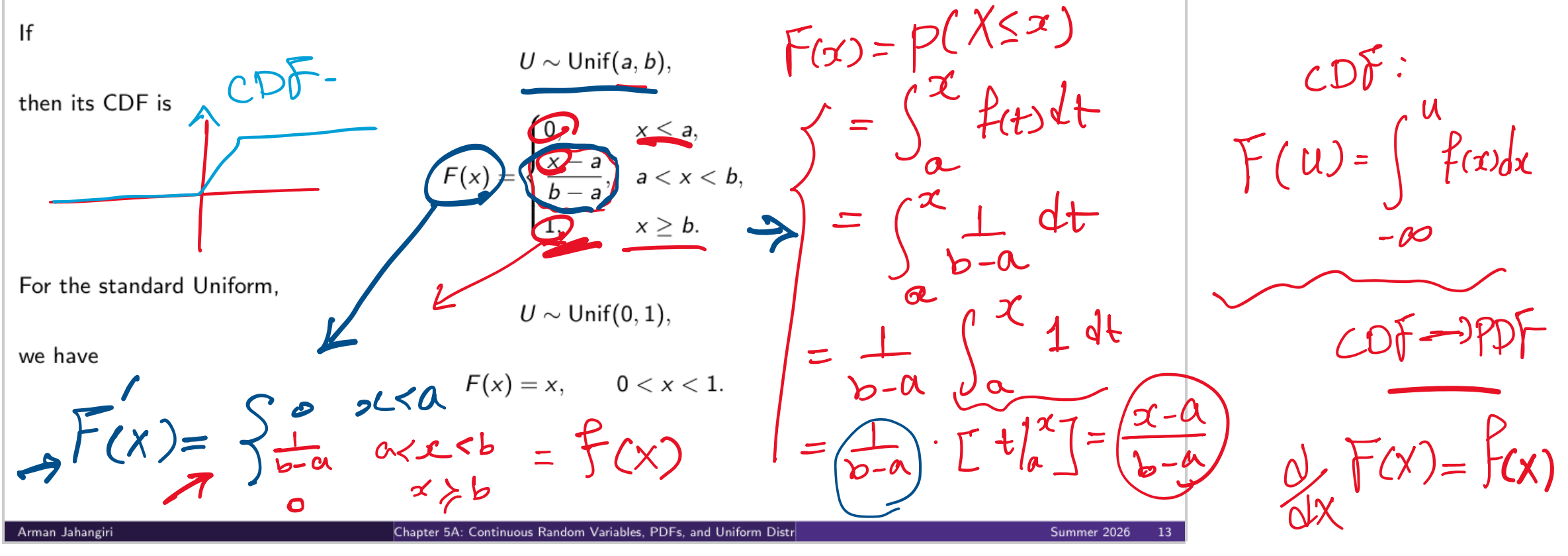
$$f(x) = \frac{1}{b-a} > 0$$

since $b > a$. And

$$\int_{-\infty}^{\infty} f(x)\,dx = \boxed{\int_a^b \frac{1}{b-a}\,dx} = \frac{1}{b-a}(b-a) = 1.$$

$$\int_a^b \frac{1}{b-a}\,dx = \frac{1}{b-a} \cdot \int_a^b dx = \frac{1}{b-a}\left(x\Big]_a^b\right) = \frac{b-a}{b-a} = 1$$

$$\boxed{\int_{-\infty}^{\infty} f(x)\,dx} = \underbrace{\int_{-\infty}^{a} 0\,dx}_{0} + \int_a^b \frac{1}{b-a}\,dx + \underbrace{\int_b^{\infty} 0\,dx}_{0}$$



Uniform CDF

If

$$U \sim \text{Unif}(a,b),$$

then its CDF is

CDF-

$$F(x) = \begin{cases} 0, & x \leq a, \\ \frac{x-a}{b-a}, & a < x < b, \\ 1, & x \geq b. \end{cases}$$

For the standard Uniform,

$$U \sim \text{Unif}(0,1),$$

we have

$$F(x) = x, \quad 0 < x < 1.$$

$$F'(x) = \begin{cases} 0 & x < a \\ \frac{1}{b-a} & a < x < b \\ 0 & x \geq b \end{cases} = f(x)$$

$$F(x) = P(X \leq x) = \int_a^x f(t)\,dt = \int_a^x \frac{1}{b-a}\,dt = \frac{1}{b-a}\int_a^x 1\,dt = \frac{1}{b-a} \cdot \left[t\Big|_a^x\right] = \frac{x-a}{b-a}$$

PDF ⇒ CDF

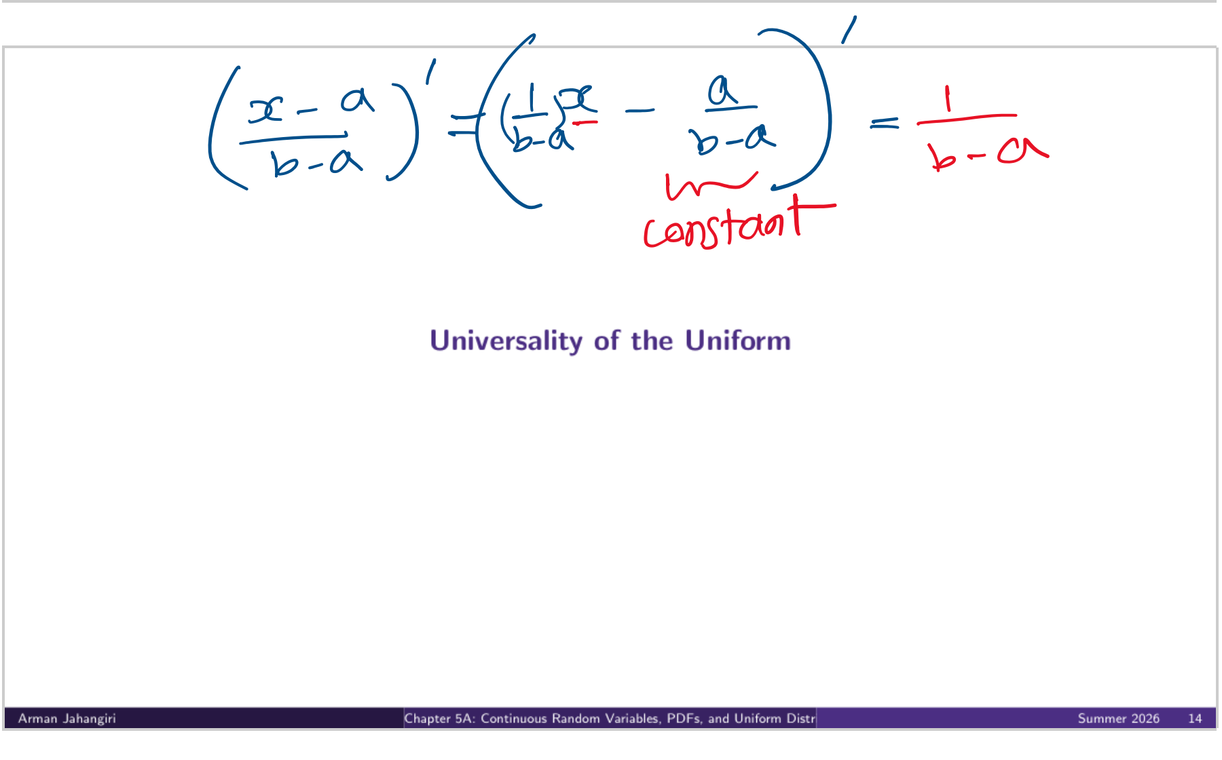
$f(x)$

CDF:

$$F(u) = \int_{-\infty}^{u} f(x)\,dx$$

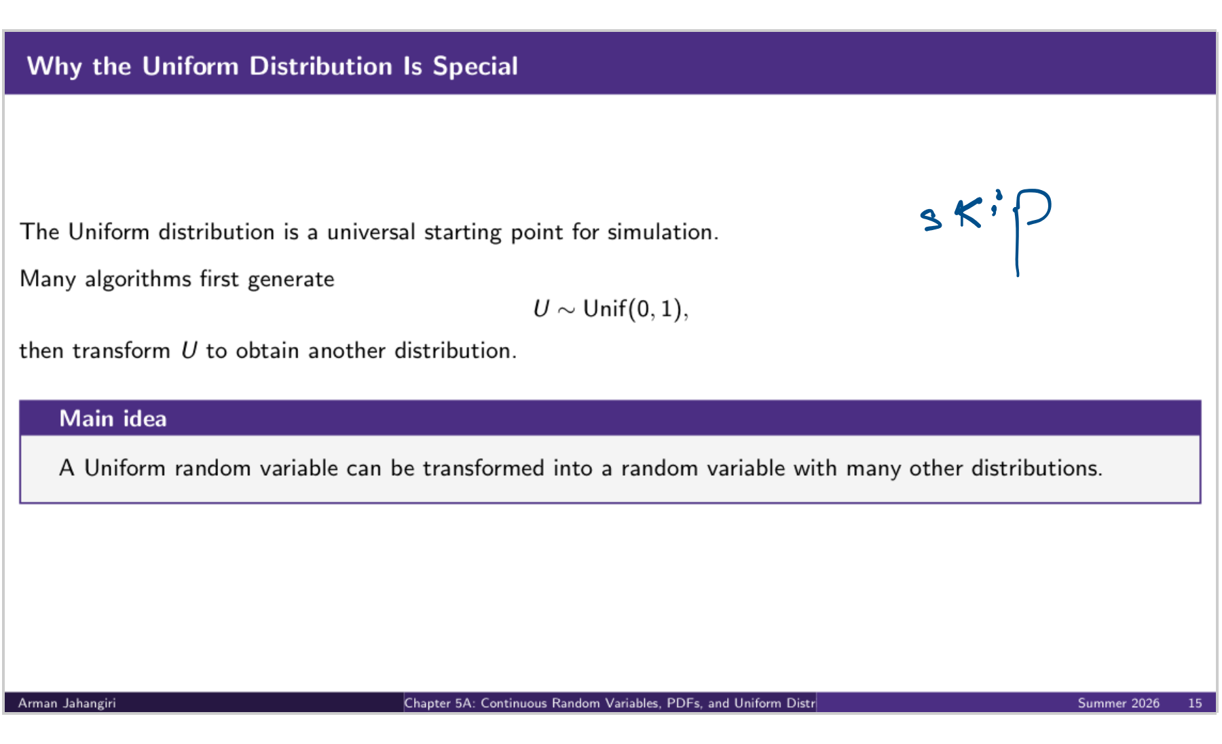
CDF ⇒ PDF

$$\frac{d}{dx}F(x) = f(x)$$



$$\left(\frac{x-a}{b-a}\right)' = \Big(\frac{1}{b-a}x - \underbrace{\frac{a}{b-a}}_{\text{constant}}\Big)' = \frac{1}{b-a}$$

Universality of the Uniform



Why the Uniform Distribution Is Special

skip

The Uniform distribution is a universal starting point for simulation.

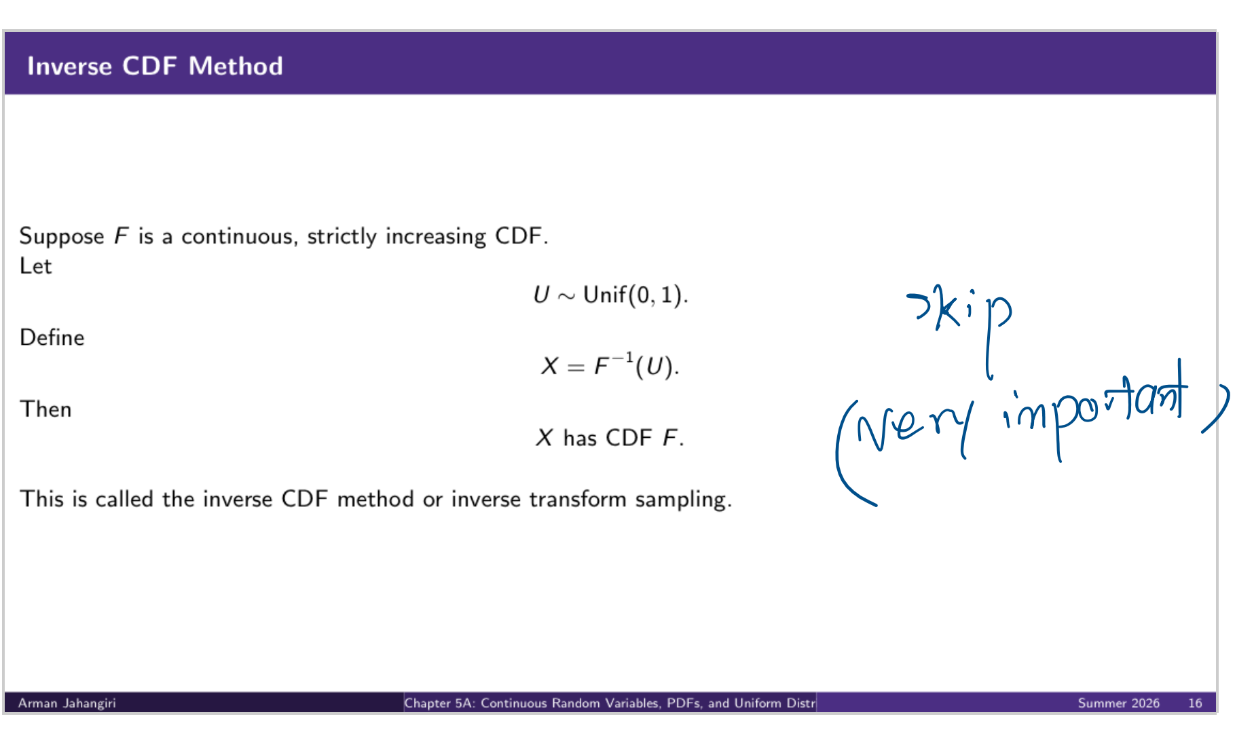
Many algorithms first generate

$$U \sim \text{Unif}(0,1),$$

then transform U to obtain another distribution.

Main idea

A Uniform random variable can be transformed into a random variable with many other distributions.



Inverse CDF Method

Suppose F is a continuous, strictly increasing CDF.
Let

$$U \sim \text{Unif}(0,1).$$

Define

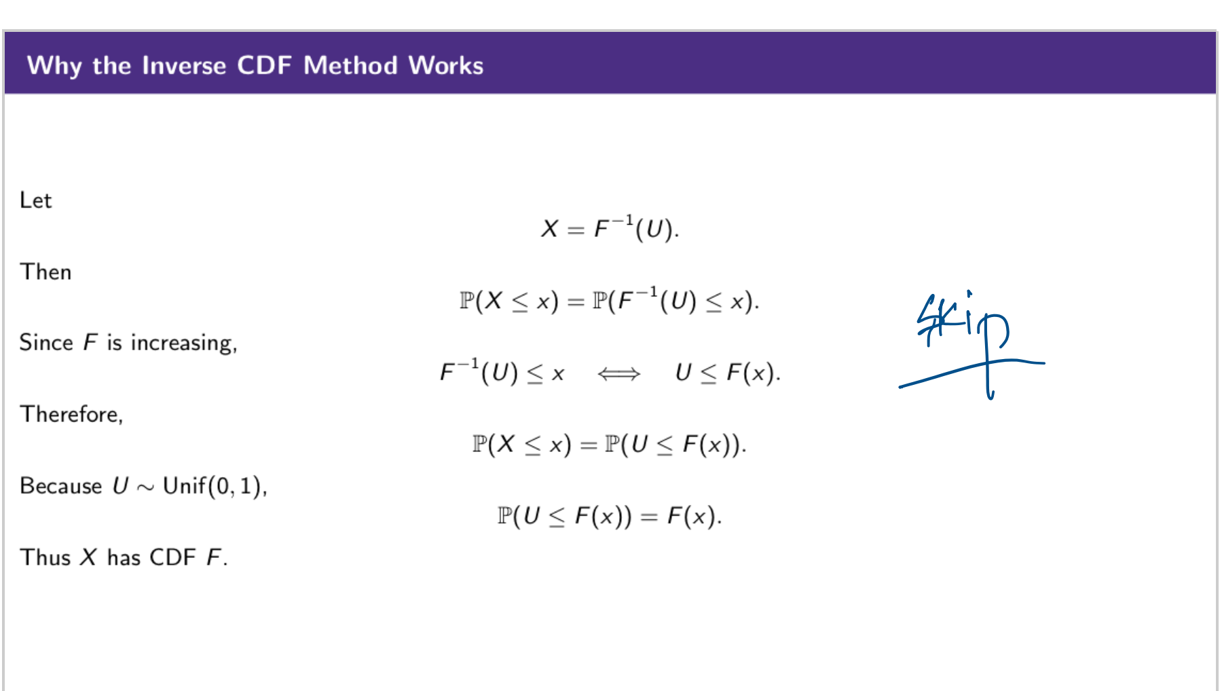
$$X = F^{-1}(U).$$

Then

$$X \text{ has CDF } F.$$

This is called the inverse CDF method or inverse transform sampling.

skip (very important)



Why the Inverse CDF Method Works

Let

$$X = F^{-1}(U).$$

Then

$$\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x).$$

Since F is increasing,

$$F^{-1}(U) \leq x \iff U \leq F(x).$$

Therefore,

$$\mathbb{P}(X \leq x) = \mathbb{P}(U \leq F(x)).$$

Because $U \sim \text{Unif}(0,1)$,

$$\mathbb{P}(U \leq F(x)) = F(x).$$

Thus X has CDF F.

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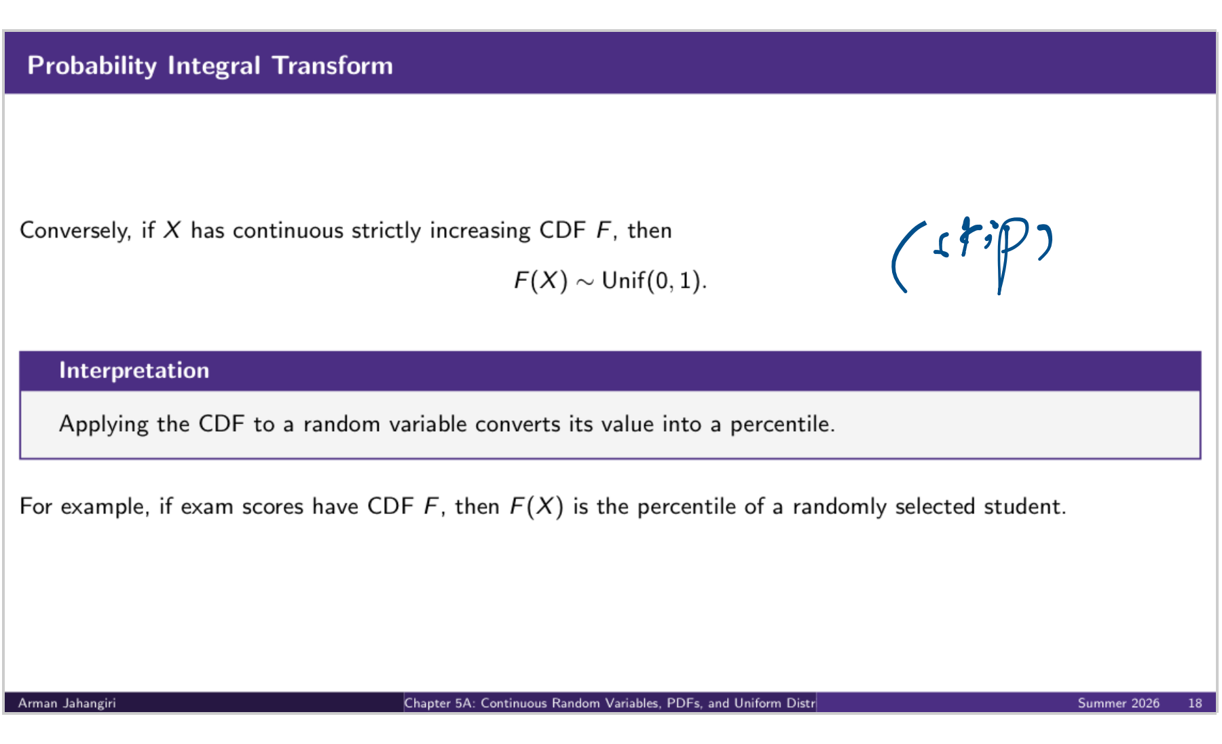


Probability Integral Transform

Conversely, if X has continuous strictly increasing CDF F, then

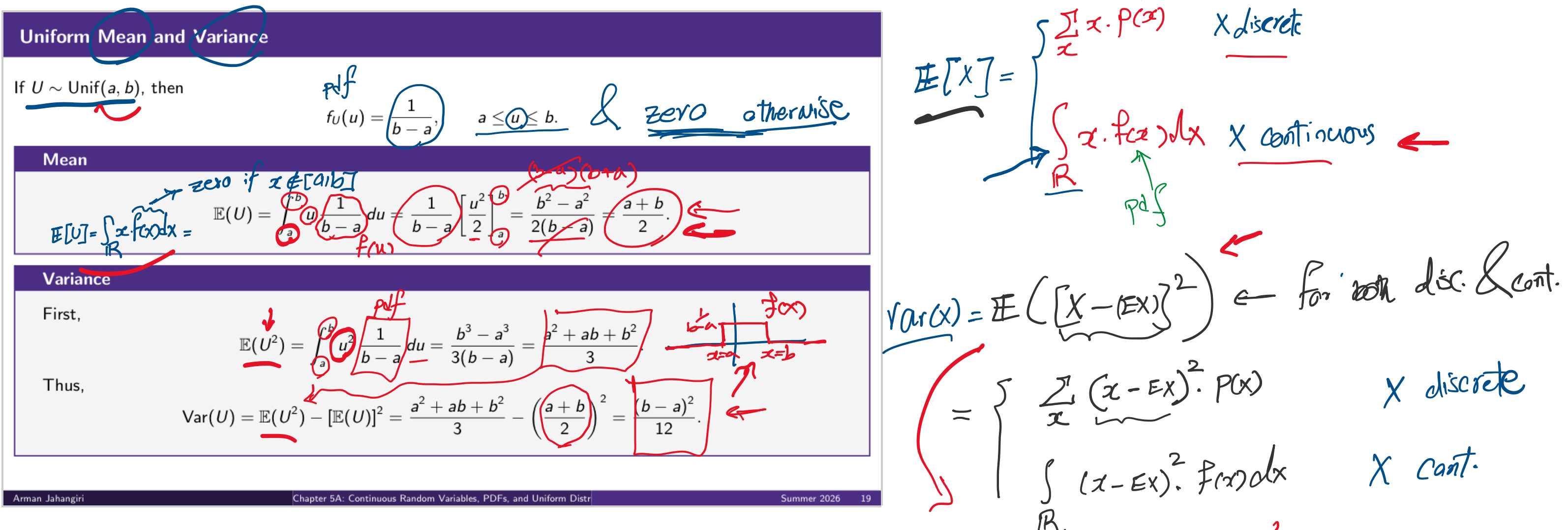
$$F(X) \sim \text{Unif}(0,1).$$

(skip)

Interpretation

Applying the CDF to a random variable converts its value into a percentile.

For example, if exam scores have CDF F, then $F(X)$ is the percentile of a randomly selected student.



Uniform Mean and Variance

If $U \sim \text{Unif}(a,b)$, then

$$\overset{\text{pdf}}{f_U(u)} = \frac{1}{b-a}, \quad a \leq u \leq b. \quad \& \text{ zero otherwise}$$

Mean

$$\mathbb{E}[U] = \int_{\mathbb{R}} x f(x)\,dx = \quad \mathbb{E}(U) = \int_a^b u \underbrace{\frac{1}{b-a}}_{f(u)}\,du = \frac{1}{b-a}\left[\frac{u^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}.$$

zero if $x \notin [a,b]$

$(b-a)(b+a)$

Variance

First,

$$\mathbb{E}(U^2) = \int_a^b u^2 \overset{\text{pdf}}{\frac{1}{b-a}}\,du = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}$$

Thus,

$$\text{Var}(U) = \mathbb{E}(U^2) - [\mathbb{E}(U)]^2 = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2 = \boxed{\frac{(b-a)^2}{12}}. \leftarrow$$

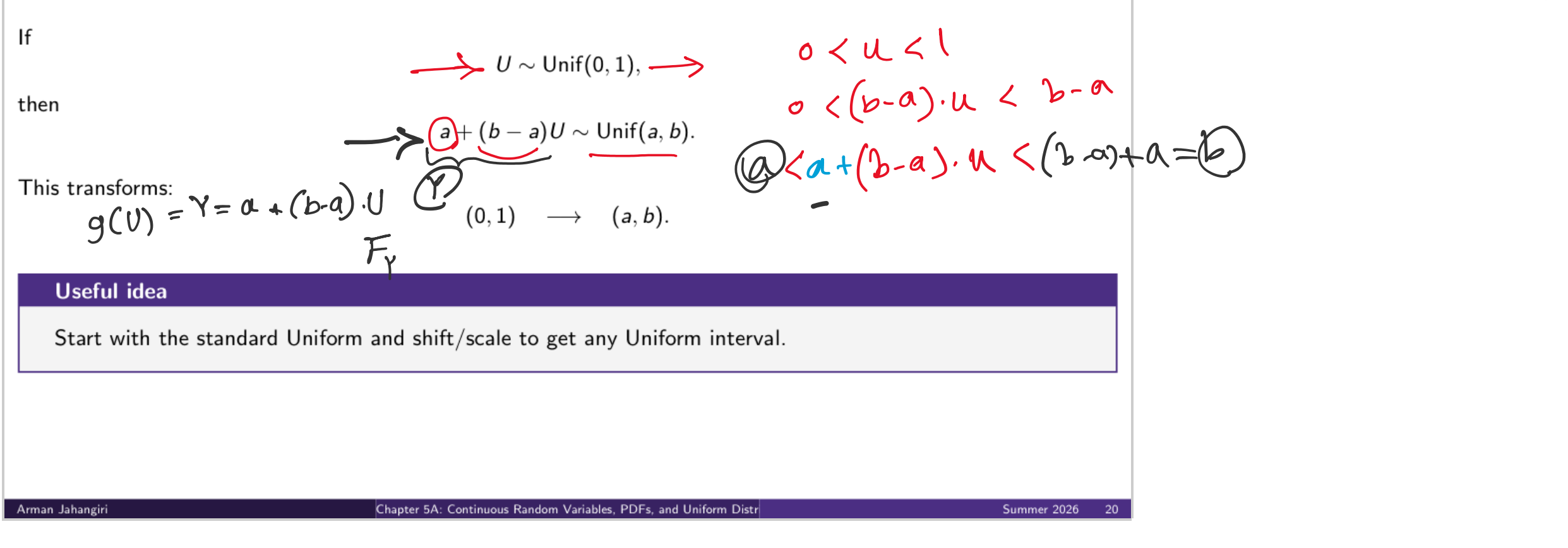
$$\mathbb{E}[X] = \begin{cases} \sum_x x \cdot p(x) & X \text{ discrete} \\ \int_{\mathbb{R}} x \cdot f(x)\,dx & X \text{ continuous} \end{cases} \leftarrow$$

PMF

pdf

$$\text{Var}(X) = \mathbb{E}\left(\left[X - \mathbb{E}X\right]^2\right) \leftarrow \text{for both disc. \& cont.}$$

$$= \begin{cases} \sum_x (x - EX)^2 \cdot p(x) & X \text{ discrete} \\ \int_{\mathbb{R}} (x - EX)^2 \cdot f(x)\,dx & X \text{ cont.} \end{cases}$$



$$\text{Var}\,X = E(X^2) - (EX)^2 \leftarrow$$

Location-Scale Transformation

If

$$\to U \sim \text{Unif}(0,1), \to$$

then

$$\to a + (b-a)U \sim \text{Unif}(a,b).$$

This transforms:

$$g(U) = Y = a + (b-a) \cdot U \quad \underset{F_Y}{Y} \quad (0,1) \longrightarrow (a,b).$$

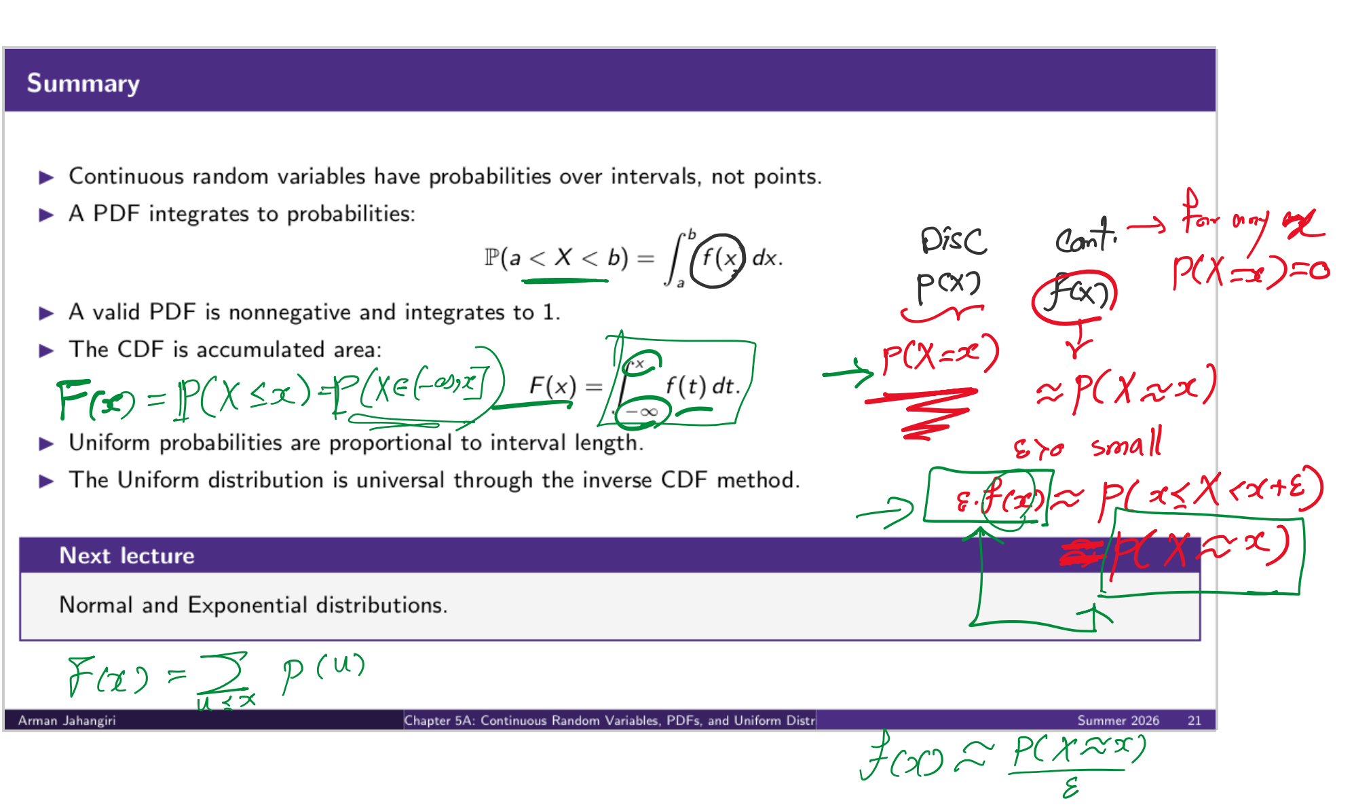
$$0 < u < 1$$

$$0 < (b-a) \cdot u < b-a$$

$$a < a + (b-a) \cdot u < (b-a) + a = b$$

Useful idea

Start with the standard Uniform and shift/scale to get any Uniform interval.



Summary

- Continuous random variables have probabilities over intervals, not points.
- A PDF integrates to probabilities:

$$\mathbb{P}(a < X < b) = \int_a^b f(x)\,dx.$$

- A valid PDF is nonnegative and integrates to 1.
- The CDF is accumulated area:

$$F(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \in (-\infty, x]) \quad F(x) = \int_{-\infty}^{x} f(t)\,dt.$$

- Uniform probabilities are proportional to interval length.
- The Uniform distribution is universal through the inverse CDF method.

Disc $\quad$ Cont. $\to$ for any x $P(X=x)=0$

$p(x) \quad f(x)$

$\to P(X=x) \quad \approx P(X \approx x)$

$\varepsilon > 0$ small

$$\to \boxed{\varepsilon \cdot f(x)} \approx P(x \leq X < x+\varepsilon) = \boxed{P(X \approx x)}$$

Next lecture

Normal and Exponential distributions.

$$F(x) = \sum_{u \leq x} p(u)$$

$$f(x) \approx \frac{P(X \approx x)}{\varepsilon}$$